

ADELSO, Sweden

WMO: 024860

Lat: 59.3579N

Lon: 17.5213E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.2	8.7	-1.0	5.2	3.6	4.5	6.9	9.8	20.8	33.9	18.6	34.0	3.0	300	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	80.6	66.1	77.4	64.3	74.5	62.7	68.4	76.5	66.4	74.4	64.6	72.1	7.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	94.2	71.8	63.5	87.6	69.6	61.5	81.7	68.3	32.6	77.1	31.0	74.6	29.7	72.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.5	14.6	-3.0	84.5	9.7	2.5	-10.0	86.3	-15.7	87.8	-21.2	89.2	-28.2	91.0		
(5)			-2.6	70.3	8.9	2.3	-9.0	71.9	-14.2	73.3	-19.2	74.5	-25.7	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	28.0	27.5	32.7	41.8	51.4	58.9	64.1	62.4	54.6	45.1	37.9	32.2	(6)
(7)		DBStd	14.44	9.01	8.91	7.97	5.87	6.06	4.99	4.79	4.51	5.33	6.14	6.34	8.51	(7)
(8)		HDD50	3271	683	629	538	253	56	3	0	0	20	174	364	551	(8)
(9)		HDD65	7450	1148	1049	1003	696	422	192	76	104	314	617	813	1016	(9)
(10)		CDD50	1377	0	0	0	7	99	268	438	385	157	22	1	0	(10)
(11)		CDD65	82	0	0	0	0	0	8	49	24	1	0	0	0	(11)
(12)		CDH74	735	0	0	0	0	21	130	437	146	1	0	0	0	(12)
(13)		CDH80	110	0	0	0	0	0	14	80	16	0	0	0	0	(13)
(14)	Wind	WSAvg	7.5	8.2	7.5	7.3	7.0	6.9	7.1	6.7	6.7	7.4	7.9	8.6	8.6	(14)
(15)	Precipitation	PrecAvg	22.2	1.5	1.2	1.1	1.3	1.5	2.4	2.8	2.7	2.1	2.0	1.9	1.8	(15)
(16)		PrecMax	25.6	2.9	2.5	2.0	3.9	3.0	4.1	5.4	6.2	6.2	3.8	4.7	4.5	(16)
(17)		PrecMin	16.9	0.3	0.2	0.1	0.2	0.8	0.9	0.4	0.1	0.4	0.9	0.5	0.4	(17)
(18)		PrecStd	1.9	0.7	0.5	0.4	0.9	0.6	0.9	1.5	1.5	1.4	0.8	1.0	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	46.6	56.4	66.7	77.0	82.3	86.1	82.1	72.9	60.8	53.1	51.7	(19)
(20)			MCWB	42.7	42.8	46.4	51.9	59.4	63.7	67.7	67.0	62.1	55.2	50.8	48.5	(20)
(21)		2%	DB	43.0	43.1	50.2	61.9	72.1	77.6	82.2	77.9	69.1	58.2	50.4	47.3	(21)
(22)			MCWB	40.3	39.5	43.0	49.6	57.9	62.8	67.1	65.6	59.8	54.3	48.3	44.7	(22)
(23)		5%	DB	40.7	40.6	46.3	57.6	67.9	74.4	78.9	74.7	65.9	56.1	48.4	44.5	(23)
(24)			MCWB	38.6	38.2	40.5	47.4	55.7	60.7	65.7	63.9	58.4	52.9	46.5	42.2	(24)
(25)		10%	DB	38.7	38.5	42.8	53.6	63.9	70.8	75.7	71.9	63.3	54.1	46.5	42.2	(25)
(26)			MCWB	36.8	36.5	38.7	44.8	53.2	59.1	64.3	62.5	57.2	51.3	44.7	40.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	43.8	46.9	53.6	62.4	66.7	71.3	71.4	63.2	57.2	51.4	48.7	(27)
(28)			MCDWB	45.5	45.8	54.5	63.3	73.8	77.4	78.8	77.9	69.8	59.1	52.8	51.2	(28)
(29)		2%	WB	40.9	40.2	43.4	50.7	59.2	64.3	69.1	67.8	61.1	54.8	48.6	45.1	(29)
(30)			MCDWB	42.7	42.2	49.1	60.3	68.7	75.6	77.8	73.9	67.0	57.4	50.2	46.9	(30)
(31)		5%	WB	38.9	38.4	41.3	47.9	56.7	62.0	67.3	65.8	59.4	53.3	46.7	42.6	(31)
(32)			MCDWB	40.6	40.2	45.5	56.2	66.1	71.2	76.1	72.0	64.2	55.6	48.1	44.0	(32)
(33)		10%	WB	36.9	36.6	39.3	45.6	54.6	60.1	65.5	64.0	58.0	51.6	45.0	40.5	(33)
(34)			MCDWB	38.5	38.4	42.3	52.5	62.7	68.6	73.8	69.7	62.4	53.8	46.3	42.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	10.5	13.4	18.3	19.0	18.7	17.9	16.5	14.1	10.1	7.0	7.3	(35)
(36)			MCDBR	8.6	9.8	17.6	26.2	26.4	26.7	23.7	21.0	17.8	11.5	8.6	7.9	(36)
(37)			MCWBR	8.2	7.7	11.7	15.2	14.6	13.7	11.2	10.6	10.3	8.7	7.8	7.3	(37)
(38)		5% WB	MCDBR	8.5	8.8	15.6	23.4	22.9	23.4	21.0	18.3	15.7	10.4	8.2	7.8	(38)
(39)			MCWBR	8.3	7.2	10.8	14.1	13.5	12.7	10.7	10.2	9.7	8.5	7.6	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.281	0.320	0.340	0.376	0.373	0.363	0.390	0.381	0.347	0.336	0.307	0.265	(40)	
(41)		taud	2.212	2.276	2.351	2.320	2.393	2.448	2.395	2.414	2.494	2.419	2.259	2.138	(41)	
(42)		Ebn at Noon	164	218	252	262	272	276	265	258	250	212	151	122	(42)	
(43)		Edh at Noon	14	22	28	34	34	32	34	30	24	19	13	10	(43)	
(44)	All-Sky Solar Radiation	RadAvg	100	300	780	1252	1628	1818	1664	1319	868	388	127	67	(44)	
(45)		RadStd	16	44	95	122	146	108	162	130	98	63	28	7	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47)	Regional (56 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	-209	-261	N/A

Nomenclature: See separate page